

# The Interaction between Competition and Environmental Policy in Polluting Industries

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- Climate change has caused various policy responses, including carbon pricing.
  - Carbon pricing in place in EU, Canada, UK, China, Japan, and many others
- Growing recognition that “green innovation” and “infrastructure investments” are key for decarbonization.
  - These and emissions are equilibrium objects.
  - This has prompted policy makers to consider whether sustainability should be part of merger policy
  - Examples: EU and Australia

**Big picture questions:** What is the impact of competition in a polluting industry? How do competition and environmental policy interact in the pursuit of decarbonization?

- The European Commission's New Draft Merger Guidelines (published April 2026):
  - In defining the key price- and non-price parameters on which firms compete, they include prices, innovation, and sustainability (among other).
  - “It [EU Merger Control] also allows for the preservation of green innovation competition, which supports sustainability and the transition to low carbon technologies.”
- The Australian Competition and Consumer Commission (ACCC) may authorize an otherwise anticompetitive merger if they pass a “net public benefit” test
  - The ACCC explicitly recognizes environmental efficiencies such as emissions reductions, renewable energy transition, and biodiversity protection as valid public benefits.
  - *Out-of-market* efficiencies are allowed.

# This Paper

We ask:

- 1 What is the equilibrium impact of competition on social welfare in the context of a polluting industry?
- 2 How does a carbon tax reshape this relationship? Does the magnitude of the tax burden matter?

Our approach:

- Oligopoly model with carbon tax
- Upstream monopolist developing an emission-abatement innovation
- The monopolist chooses how many licenses of its technology it sells

# Model Setup

## Market Structure

- $n$  firms competing à la Cournot.
- Inverse demand:  $P(Q) = 1 - Q$ .
- Baseline carbon tax:  $\tau < 1$ .

## Technology & Abatement

- An incremental abatement technology is available.
- Developed by a monopolist innovator choosing how many licenses to sell
  - Today: The magnitude of the innovation  $\alpha$  is exogenous
- $m$  adopters, each with emissions  $(1 - \alpha)q_i$ .
- $m - n$  non-adopters, each with emissions:  $q_i$
- Adopters cut their tax bill (per unit produced),  $C_A(q_i) = (1 - \alpha)\tau q_i$ , relative to non-adopters,  $C_N(q_i) = \tau q_i$ .

## Environmental Externality:

- Social cost of carbon:  $c$  per unit of emissions

# Oligopoly Equilibrium (Given $n$ and $m$ )

Under the incremental innovation assumption, both types of firms produce positive quantities.

## Equilibrium Quantities:

- Adopters:  $q_A = \frac{1-\tau+(n-m+1)\alpha\tau}{n+1}$
- Non-Adopters:  $q_N = \frac{1-\tau-m\alpha\tau}{n+1}$
- Total Industry:  $Q = \frac{n(1-\tau)+m\alpha\tau}{n+1}$

## Equilibrium Profits:

- Adopters:  $\pi_A = (q_A)^2$
- Non-Adopters:  $\pi_N = (q_N)^2$

# The Innovator's Problem

The innovator sets  $m$  to maximize licensing revenue:  $\Pi(m) = m\Delta(m, n)$ . The incremental rent (willingness to pay) is  $\Delta(m, n) = \pi_A(m, n) - \pi_N(m - 1, n)$ .

The optimal licensing strategy depends strictly on the size of the carbon tax relative to a threshold,  $t_\alpha = \frac{2}{2+\alpha(3n-2)}$ :

## Small Tax ( $\tau \leq t_\alpha$ )

The innovator sells to everyone:  $\mathbf{m}^* = \mathbf{n}$ .

## Large Tax ( $\tau > t_\alpha$ )

The innovator restricts licenses to create scarcity:  $\mathbf{m}^* = \frac{n}{4} + \frac{1-(1-\alpha)\tau}{2\alpha\tau}$ .

# Comparative Statics for Optimal Licenses

Focusing on the restricted licensing regime (Large Tax,  $\tau > t_\alpha$ ):

- **Market Size ( $n$ ):**  $\frac{\partial m^*}{\partial n} = \frac{1}{4} > 0$ .
  - For every 4 new firms entering the market, the innovator only issues 1 additional license.
- **Abatement Effectiveness ( $\alpha$ ):**  $\frac{\partial m^*}{\partial \alpha} = -\frac{(1-\tau)}{2\alpha^2\tau} < 0$ .
  - The larger the abatement potential, the fewer licenses sold (to maintain a higher premium per license).
- **Carbon Tax ( $\tau$ ):**  $\frac{\partial m^*}{\partial \tau} = -\frac{1}{2\alpha\tau^2} < 0$ .
  - A higher carbon tax incentivizes the innovator to sell fewer licenses.

# Deconstructing Total Welfare

Total Welfare ( $W$ ) captures a tension between three economic forces.

$$W = \underbrace{TS}_{\text{Total Surplus}} - \underbrace{c \times E}_{\text{Environmental Externality}}$$

- 1 **Total Surplus ( $TS = CS + PS$ ):**  $Q - Q^2/2$ . The overall value generated by the market, driven entirely by total quantity  $Q$ .
- 2 **Environmental Externality ( $c \times E$ ):**  $c \times [(1 - \alpha)mq_A + (n - m)q_N]$ . The aggregate social cost of production, where  $c$  is the social cost of carbon
- 3 **Composition Effect:** The technological makeup of the market (dictated by the number of licenses  $m$ ), which determines whether production is clean or dirty.

# The Fundamental Tension: Surplus vs. Externalities

Before considering the innovator's licensing strategy, increasing market competition ( $n$ ) creates a well understood tension (Buchanan 1969):

## 1. Total Surplus Increases (Positive for Welfare)

As  $n$  increases, total quantity  $Q$  strictly increases. Because  $TS$  is strictly increasing in  $Q$ , more competition unambiguously increases the Total Surplus of the market.

## 2. Environmental Externality Increases (Negative for Welfare)

Because emissions are a byproduct of production, the total environmental damage scales with  $Q$ . Every additional unit generated by competition pumps more emissions into the atmosphere.

# New Effect: The Composition Effect

**Composition Effect:** what kind of firm is entering the market when  $n$  increases?

This is entirely dictated by the innovator's licensing strategy ( $m^*$ ):

- If the innovator grants licenses freely ( $m = n$ ), the market composition remains **“clean”**. The emissions per unit produced stay low, easing the externality tension.
- If the innovator restricts licenses ( $m < n$ ), the market composition turns **“dirty”**. The marginal entrant is forced to produce without abatement, causing the environmental externality to accelerate rapidly.

# Welfare Synthesis: The Small Tax Regime ( $\tau \leq t_\alpha$ )

**Market State:** The tax is low enough that the innovator sells to everyone ( $m = n$ ).

**The Economic Balance as  $n \uparrow$ :**

- **Total Surplus:** Increases as competition drives up total quantity  $Q$ .
- **Composition Effect:** Because every firm is clean, the **Environmental Externality** of the expanded production is heavily discounted by the abatement factor  $(1 - \alpha)$ .

# Welfare Synthesis: The Large Tax Regime ( $\tau > t_\alpha$ )

**Market State:** The tax is high, so the innovator creates an artificial bottleneck, granting only 1 license for every 4 new firms.

**The Economic Balance as  $n \uparrow$ :**

- **Total Surplus:** Continues to rise with  $Q$ , benefiting the market.
- **Composition Effect:** The market floods with dirty production. The marginal entrants lack the technology, causing the **Environmental Externality** to explode.

# Visualizing the Inflection Point

Let's visualize these effects with specific parameters:

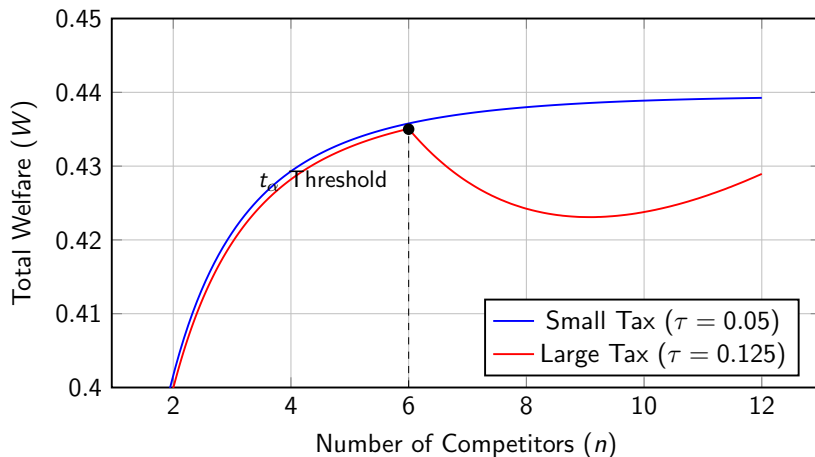
## Parameters:

- Abatement Effectiveness ( $\alpha$ ): 0.875 (87.5% reduction)
- Social Cost of Carbon ( $c$ ): 0.5

We compare two scenarios:

- 1 **Small Tax** ( $\tau = 0.05$ ): The market stays in the full-adoption regime.
- 2 **Large Tax** ( $\tau = 0.125$ ): The market crosses the licensing threshold exactly at  $n = 6$ .

# Total Welfare vs. Market Competition ( $n$ )



# Moderate Abatement: The Overproduction Problem

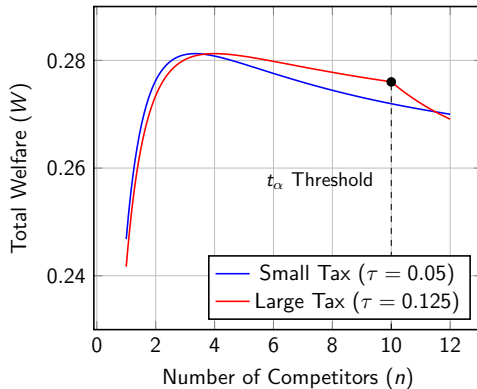
## Parameters:

- Abatement:  $\alpha = 0.5$  (Moderate)
- Social Cost of Carbon:  $c = 0.5$

## The Dynamics Swap:

Because the technology is less effective, the socially optimal total quantity drops.

As  $n$  grows, the market begins to overproduce a dirty good.



# Concluding Remarks

- Competition and the carbon tax interact non-trivially in equilibrium
- The Bottleneck Effect: When the tax is too high, introducing new competitors to the market primarily introduces heavy polluters, leading to a net decrease in societal welfare.
- Next steps:
  - Endogenous  $\alpha$
  - Radical vs incremental innovation
  - Optimal policy design